exciting science fiction, elbow room that Christopher Nolan exploits with great finesse in *Interstellar*; see Chapters 28–31.

Truth, Educated Guesses, and Speculations

The science of *Interstellar* lies in all four domains: Newtonian, relativistic, quantum, and quantum gravity. Correspondingly, some of the science is known to be true, some is an educated guess, and some is speculation.

To be *true*, the science must be based on well-established physical laws (Newtonian, relativistic, or quantum), and it must have enough basis in observation that we are confident of how to apply the well-established laws.

In precisely this sense, neutron stars and their magnetic fields, as described in Chapter 2, are true. Why? First, neutron stars are firmly predicted to exist by the quantum and relativistic laws. Second, astronomers have studied in enormous detail the pulsar radiation from neutron stars (pulses of light, X-rays, and radio waves described in Chapter 2). These pulsar observations are beautifully and accurately explained by the quantum and relativistic laws, if the pulsar is a spinning neutron star; and no other explanation has ever been found. Third, neutron stars are firmly predicted to form in astronomical explosions called supernovae, and pulsars are seen at the centers of big, expanding gas clouds, the remnants of old supernovae. Thus, we astrophysicists have no doubt; neutron stars really do exist and they really do produce the observed pulsar radiation.

Another example of a truth is the black hole Gargantua and the bending of light rays by which it distorts images of stars (Figure 3.3). Physicists call this distortion “gravitational lensing” because it is similar